

Telco Customer Churn Analysis

This is a summary of the churn analysis notebook. The dataset has 7,043 customers and 21 columns covering demographics, account details, subscribed services, billing, and contract type. The data was clean to start with — no missing values and no duplicate customer IDs, aside from one small fix (TotalCharges had blank entries for new customers, which were set to 0).

Executive Summary

About 1 in 4 customers churned — the overall churn rate is 26.54%. Digging into why, a few patterns stand out clearly: customers on month-to-month contracts churn a lot more than those on 1- or 2-year contracts, new customers (first 1-2 months) are the most likely to leave, people paying by electronic check churn more than those on auto-pay, and customers without add-ons like Online Security or Tech Support are more likely to leave. Senior citizens also churn at a higher rate than non-seniors. Gender doesn't seem to matter at all.

Quick Numbers

- Total customers: 7,043
- Overall churn rate: 26.54%
- Average tenure: 32.4 months
- Average monthly charges: \$64.76
- Missing values: none
- Duplicate customer IDs: none

What Was Done in the Notebook

- Loaded the data and checked its structure (rows, columns, data types).
- Cleaned TotalCharges (it had blanks stored as text) and confirmed no nulls or duplicates.
- Converted SeniorCitizen from 0/1 to No/Yes so it's easier to read in charts.
- Looked at overall churn split with a count plot and a pie chart (26.54% churned).
- Checked churn by gender (no real difference) and by senior citizen status (seniors churn more).
- Plotted tenure vs churn — churn is highest in the first 1-2 months, then drops off a lot.
- Compared churn by contract type — month-to-month customers churn far more than yearly contracts.
- Looked at churn across service add-ons (Online Security, Tech Support, Online Backup, streaming, etc.) — customers without these tend to churn more.
- Checked churn by payment method — electronic check users churn the most.

Key Findings

- Contract type is the biggest factor — month-to-month = high churn risk.
- New customers (under 2 months) are the most likely to leave.
- Electronic check payers churn more than people on auto-pay.
- Customers without Online Security / Tech Support / Online Backup churn more.
- Senior citizens churn at a higher rate proportionally.
- Fiber optic internet customers churn more than DSL customers.
- Gender has no real effect on churn.

Recommendations

- Encourage month-to-month customers to switch to 1- or 2-year contracts with a discount or perk.
- Focus retention efforts on the first 60 days — that's when most churn happens.
- Push people toward auto-pay instead of electronic check.
- Bundle in Online Security and Tech Support for new customers to increase stickiness.
- Use these factors (contract, tenure, payment method, add-ons) to build a simple churn-risk score for targeting at-risk customers.

Tools Used

Python, pandas for data cleaning, and Matplotlib/Seaborn for the charts (count plots, a pie chart, a stacked bar chart, and a tenure histogram).